Linear Algebra – MAT 2610

Section 1.1

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What is a Linear Equation?

$$a_1 x_1 + a_2 x_2 + \cdots + a_n x_n = b$$

where b and the coefficients $a_1, \ldots, a_n$ are real or complex numbers (we will only use real numbers in this course)

$$3x_1 + 4x_2 - 3x_3 + \sqrt{41}\, x_4 = 27/17$$

$$3x_1 = 2x_4 - 27 + 9x_3 + x_2$$

A System of Linear Equations

This is a collection of one or more linear equations involving the same variables.

$$\begin{cases} 2x_1 + 3x_2 + 7x_3 = 18 \\ -2x_1 + 2x_3 = 0 \\ x_2 = 3 \end{cases}$$

System with 3 equations in 3 variables

A Solution to a System of Linear Equations

This is a list $(s_1, s_2, \ldots, s_n)$ of numbers that makes each equation a true statement when substituted for the variables $(x_1, x_2, \ldots, x_n)$ respectively.

$$
\begin{aligned}
2x_1 + 3x_2 + 7x_3 &= 18 \\
-2x_1 + 2x_3 &= 0 \\
x_2 &= 3
\end{aligned}
$$

$(1,3,1)$ substituted for (x_1, x_2, x_3) makes each equation true.

$2(1) + 3(3) + 7(1) = 2 + 9 + 7 = 18$ ✓

$-2(1) + 2(1) = -2 + 2 = 0$ ✓

A Solution Set

The collection of all possible solutions to a linear system is called a **solution set**.

Two linear systems are **equivalent** if they have the same solution set.

Solution Set Examples

$$2x_1 + 3x_2 = 11$$
$$-2x_1 + 2x_2 = -6$$

Has a single solution $(x_1, x_2) = (4,1)$.

$2(4) + 3(1) = 8 + 3 = 11$ ✓

$-2(4) + 2(1) = -8 + 2 = -6$ ✓

Solution Set Examples

$$x_1 + 4x_2 = 8$$
$$2x_1 - x_2 = 7$$

Also has a single solution $(x_1, x_2) = (4,1)$.

$(4) + 4(1) = 4 + 4 = 8$ ✓

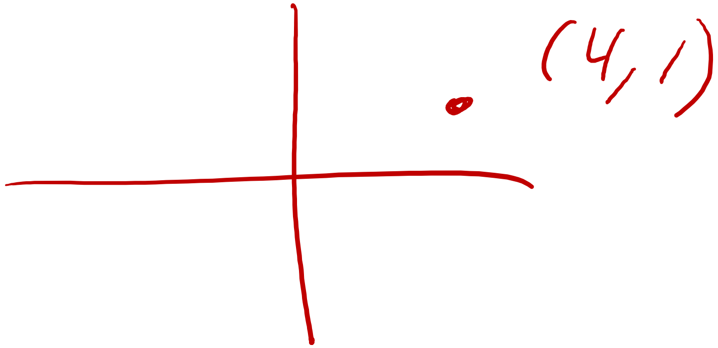
$2(4) - (1) = 8 - 1 = 7$ ✓

Solution Set Examples

Notice that these two systems of linear equations

$$\begin{aligned} 2x_1 + 3x_2 &= 11 \\ -2x_1 + 2x_2 &= -6 \end{aligned} \qquad \longrightarrow \qquad \begin{aligned} x_1 + 4x_2 &= 8 \\ 2x_1 - x_2 &= 7 \end{aligned}$$

are quite different but both have the same solution set $\boxed{(x_1, x_2) = (4,1)}$. These two systems are equivalent.



Consistent

A system of linear equations is said to be **consistent** if it has at least one solution.

If not, it is **inconsistent**.

A system of linear equations has

1. No solution, or — inconsist
2. Exactly one solution, or
3. Infintely many solutions.

consistent

Matrix

A **matrix** is a rectangular array of numbers

A matrix with *m* rows and *n* columns is referred to as an **m by n matrix** (or **m x n matrix**)

We say that an $m \times n$ matrix has **size** m by n

Note that the number of rows is <u>always</u> given first

$\begin{bmatrix} 3 & 3 \\ -2 & 2 \\ 3 & 1 \end{bmatrix}$ is a 3 X 2 matrix

$\begin{bmatrix} 4 & -2 & -2 \\ 5 & -1 & 3 \end{bmatrix}$ is a 2 X 3 matrix

Matrix Notation

If we have:

$$x_1 + 3x_2 + 2x_3 = 13$$
$$2x_1 + 3x_2 - x_3 = 5$$
$$2x_1 - x_3 = 5$$

$$\begin{array}{ccc} x_1 & x_2 & x_3 \end{array}$$
$$\begin{bmatrix} 1 & 3 & 2 \\ 2 & 3 & -1 \\ 2 & 0 & -1 \end{bmatrix}$$

The matrix $\begin{bmatrix} 1 & 3 & 2 \\ 2 & 3 & -1 \\ 2 & 0 & -1 \end{bmatrix}$ is called the **coefficient matrix.**

If we add in another column, consisting of the constants, we get the

matrix $\left[\begin{array}{ccc|c} 1 & 3 & 2 & 13 \\ 2 & 3 & -1 & 5 \\ 2 & 0 & -1 & 5 \end{array}\right]$. This is the **augmented matrix.**

Solving The System

$$x_1 + 3x_2 + 2x_3 = 13$$
$$2x_1 + 3x_2 - x_3 = 5$$
$$2x_1 + x_3 = 5$$

$$\begin{bmatrix} 1 & 3 & 2 & 13 \\ 2 & 3 & -1 & 5 \\ 2 & 0 & 1 & 5 \end{bmatrix}$$

Eliminate x_1 from the second equation using the first

$$x_1 + 3x_2 + 2x_3 = 13 \xrightarrow{\times 2} 2x_1 + 6x_2 + 4x_3 = 26$$
$$2x_1 + 3x_2 - x_3 = 5 \longrightarrow [-]\ 2x_1 + 3x_2 - x_3 = 5$$

$$3x_2 + 5x_3 = 21$$

Solving The System (2)

$$x_1 + 3x_2 + 2x_3 = 13$$
$$3x_2 + 5x_3 = 21$$
$$2x_1 + x_3 = 5$$

$$\begin{bmatrix} 1 & 3 & 2 & 13 \\ 0 & 3 & 5 & 21 \\ 2 & 0 & 1 & 5 \end{bmatrix}$$

Eliminate x_1 from the third equation using the first

$$x_1 + 3x_2 + 2x_3 = 13 \xrightarrow{\times 2} 2x_1 + 6x_2 + 4x_3 = 26$$
$$2x_1 + x_3 = 5 \longrightarrow 2x_1 + x_3 = 5$$

$$6x_2 + 3x_3 = 21$$

Solving The System (3)

$$x_1 + 3x_2 + 2x_3 = 13$$
$$3x_2 + 5x_3 = 21$$
$$6x_2 + 3x_3 = 21$$

$$\begin{bmatrix} 1 & 3 & 2 & 13 \\ 0 & 3 & 5 & 21 \\ 0 & 6 & 3 & 21 \end{bmatrix}$$

Eliminate x_2 from the third equation using the second

$$3x_2 + 5x_3 = 21 \xrightarrow{\times 2} 6x_2 + 10x_3 = 42$$
$$6x_2 + 3x_3 = 21 \longrightarrow 6x_2 + 3x_3 = 21$$

$$7x_3 = 21$$

Solving The System (4)

$$x_1 + 3x_2 + 2x_3 = 13$$
$$3x_2 + 5x_3 = 21$$
$$7x_3 = 21$$

$$\longrightarrow \begin{bmatrix} 1 & 3 & 2 & 13 \\ 0 & 3 & 5 & 21 \\ 0 & 0 & 7 & 21 \end{bmatrix}$$

Now notice we can easily find the value for x_3

$$7x_3 = 21$$

$$x_3 = 3$$

Solving The System (5)

$$x_1 + 3x_2 + 2x_3 = 13$$
$$3x_2 + 5x_3 = 21$$
$$x_3 = 3$$

$$\begin{bmatrix} 1 & 3 & 2 & 13 \\ 0 & 5 & 5 & 21 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

Using the value for x_3 we can solve for x_2

$$3x_2 + 5(3) = 21$$
$$3x_2 + 15 = 21$$
$$3x_2 = 6$$
$$x_2 = 2$$

Solving The System (6)

$$x_1 + 3x_2 + 2x_3 = 13$$
$$x_2 = 2$$
$$x_3 = 3$$

$$\begin{bmatrix} 1 & 3 & 2 & 13 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

Using the value for x_2 and x_3 we can solve for x_1

$$x_1 + 3(2) + 2(3) = 13$$
$$x_1 + 6 + 6 = 13$$
$$x_1 + 12 = 13$$
$$x_1 = 1$$

Solving The System (7)

$$x_1 = 1$$
$$x_2 = 2$$
$$x_3 = 3$$

$$\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

Check your work !!

$$x_1 + 3x_2 + 2x_3 = 13$$
$$2x_1 + 3x_2 - x_3 = 5$$
$$2x_1 + x_3 = 5$$

$1 + 3(2) + 2(3) = 1 + 6 + 6 = 13$ ✓

$2(1) + 3(2) - (3) = 2 + 6 - 3 = 5$ ✓

$2(1) + (3) = 5$ ✓

Elementary Row Operations

1. Add a multiple of one row to another row:

$$\begin{bmatrix} 1 & 3 & 2 \\ 2 & 1 & -1 \\ 2 & 0 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 3 & 2 \\ 4 & 7 & 3 \\ 2 & 0 & 3 \end{bmatrix} \ (2r_1 + r_2 \rightarrow r_2)$$

2. Interchange any two rows of the matrix:

$$\begin{bmatrix} 1 & 3 & 2 \\ 2 & 1 & -1 \\ 2 & 0 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 3 & 2 \\ 2 & 0 & 3 \\ 2 & 1 & -1 \end{bmatrix} \ (r_2 \leftrightarrow r_3)$$

3. Multiply every entry of some row by some nonzero scalar:

$$\begin{bmatrix} 1 & 3 & 2 \\ 2 & 1 & -1 \\ 2 & 0 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 3 & 2 \\ 6 & 3 & -3 \\ 2 & 0 & 3 \end{bmatrix} \ (3r_2 \rightarrow r_2)$$

Elementary Row Operations (2)

Any to matrices are called **row equivalent** if there is a sequence of elementary row operations that transforms one matrix into the other.

$$\begin{bmatrix} 1 & 3 & 2 & 13 \\ 2 & 3 & -1 & 5 \\ 2 & 0 & 1 & 5 \end{bmatrix} \text{ is row equivalent to } \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

$x_1 + 0x_2 + 0x_3 = 1$

$0x_1 + x_2 + 0x_3 = 2$

$0x_1 + 0x_2 + x_3 = 3$

It is important to note that elementary row operations are reversible.

If the augmented matrices of two linear systems are row equivalent, then the two systems have the same solution set.

Example:

$$\begin{aligned} 2x + y - z &= 2 \\ x + 3y + 2z &= 1 \\ x + y - z &= 2 \end{aligned}$$

Solve

$$\begin{array}{c} \begin{array}{cccc} x & y & z & \text{const} \end{array} \\ \begin{bmatrix} 2 & 1 & -1 & 2 \\ 1 & 3 & 2 & 1 \\ 1 & 1 & -1 & 2 \end{bmatrix} \end{array}$$

x_1

$$\begin{bmatrix} 2 & 1 & -1 & 2 \\ 1 & 3 & 2 & 1 \\ 1 & 1 & -1 & 2 \end{bmatrix} \xrightarrow{r_1 - 2r_2 \to r_2} \begin{bmatrix} 2 & 1 & -1 & 2 \\ 2-2(1) & 1-2(3) & -1-2(2) & 2-2(1) \\ 1 & 1 & -1 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 1 & -1 & 2 \\ 0 & -5 & -5 & 0 \\ 1 & 1 & -1 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 1 & -1 & 2 \\ 0 & -5 & -5 & 0 \\ 1 & 1 & -1 & 2 \end{bmatrix} \xrightarrow{r_1 - 2r_3 \to r_3} \begin{bmatrix} 2 & 1 & -1 & 2 \\ 0 & -5 & -5 & 0 \\ 0 & -1 & 1 & -2 \end{bmatrix} \xrightarrow{r_2 - 5r_3 \to r_3} \begin{bmatrix} 2 & 1 & -1 & 2 \\ 0 & -5 & -5 & 0 \\ 0 & 0 & -10 & 10 \end{bmatrix}$$

$$\begin{array}{cccc} x & y & z & \text{const} \end{array}$$
$$\begin{bmatrix} 2 & 1 & -1 & 2 \\ 0 & -5 & -5 & 0 \\ 0 & 0 & -10 & 10 \end{bmatrix} \longrightarrow$$

$$2x + y - z = 2$$
$$-5y - 5z = 0$$
$$-10z = 10 \to z = -1$$

$$-5y - 5(-1) = 0$$
$$-5y + 5 = 0$$
$$y = 1$$

$$2x + (1) - (-1) = 2$$
$$2x + 2 = 2$$
$$x = 0$$

Example:

$$3x + 2y = 5$$
$$6x + 4y = 8$$

Solve

$$\begin{bmatrix} 3 & 2 & 5 \\ 6 & 4 & 8 \end{bmatrix}$$

$$\begin{bmatrix}3 & 2 & 5\\6 & 4 & 8\end{bmatrix} \xrightarrow{2r_1 - r_2 \to r_2} \begin{array}{c} \begin{matrix} x & y & \text{const} \end{matrix} \\ \begin{bmatrix}3 & 2 & 5\\0 & 0 & 2\end{bmatrix} \end{array}$$

$$\begin{aligned} 3x + 2y &= 5 \\ 0x + 0y &= 2 \end{aligned} \Rightarrow 0 = 2$$